

Customer Segmentation

Unsupervised Learning: RFM Features, PCA, and K-Means Clustering

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1 OBJECTIVE

Segment customers into behaviorally distinct groups using unsupervised learning, so the business can target each segment with tailored marketing actions such as loyalty programs, win-back campaigns, and discount targeting.

AT A GLANCE

- Three customer segments identified from twelve RFM and behavioral features using PCA plus K-Means ($K = 3$).
- Segment sizes: 63.9% core, 35.4% high value, 0.7% highly engaged niche.
- Average silhouette score of 0.272, indicating moderate but usable cluster separation.

2 METHODOLOGY

2.1 Customer-Level Aggregation

Order-level data was aggregated to one row per CustomerID, computing RFM-style and behavioral features: Recency, Frequency, Monetary, AvgOrderValue, AvgQuantity, AvgItemsInCart, DiscountRate, DeliveryRate, and UniqueProducts, plus derived ratios MonetaryPerVisit, ItemsPerOrder, and AvgPricePerItem. Twelve features were used in total.

2.2 Standardization and Dimensionality Reduction

All twelve features were standardized (mean 0, standard deviation 1) via StandardScaler, then reduced with Principal Component Analysis (PCA). Enough components were retained to explain at least 95 percent of variance for clustering; a separate two-component PCA projection was created purely for visualization.

2.3 Choosing K

K-Means was fit for K equal to 2 through 10 on the PCA-reduced data. Both the elbow method (within-cluster sum of squares) and the silhouette score were used to select the optimal number of clusters.

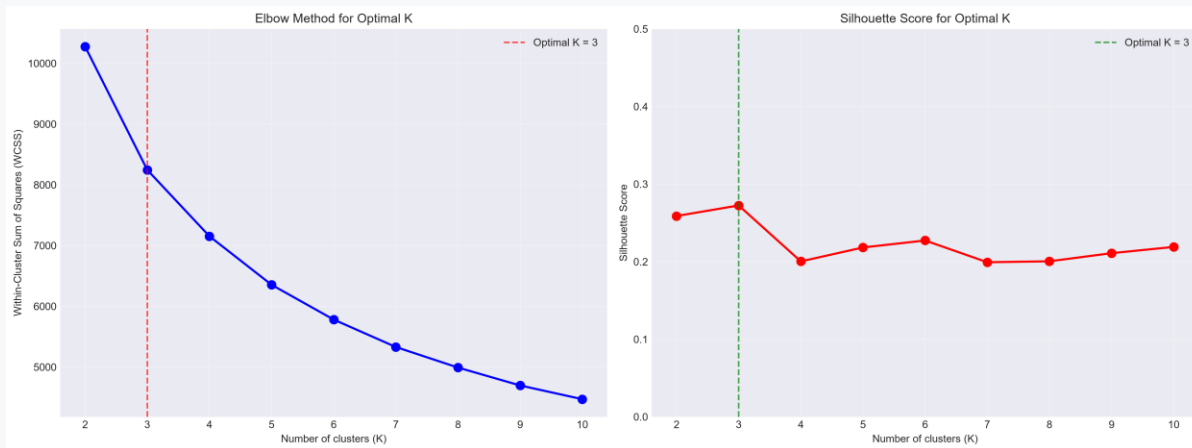


FIGURE 1. Elbow method (left) and silhouette score (right) across $K = 2$ to 10 ; $K = 3$ was selected as optimal.

2.4 Final Clustering ($K = 3$)

K-Means was re-fit with $K = 3$ on the PCA-reduced feature space. Cluster centroids were reverse-transformed back into the original feature units (inverse PCA, then inverse scaling) for interpretable business reporting.

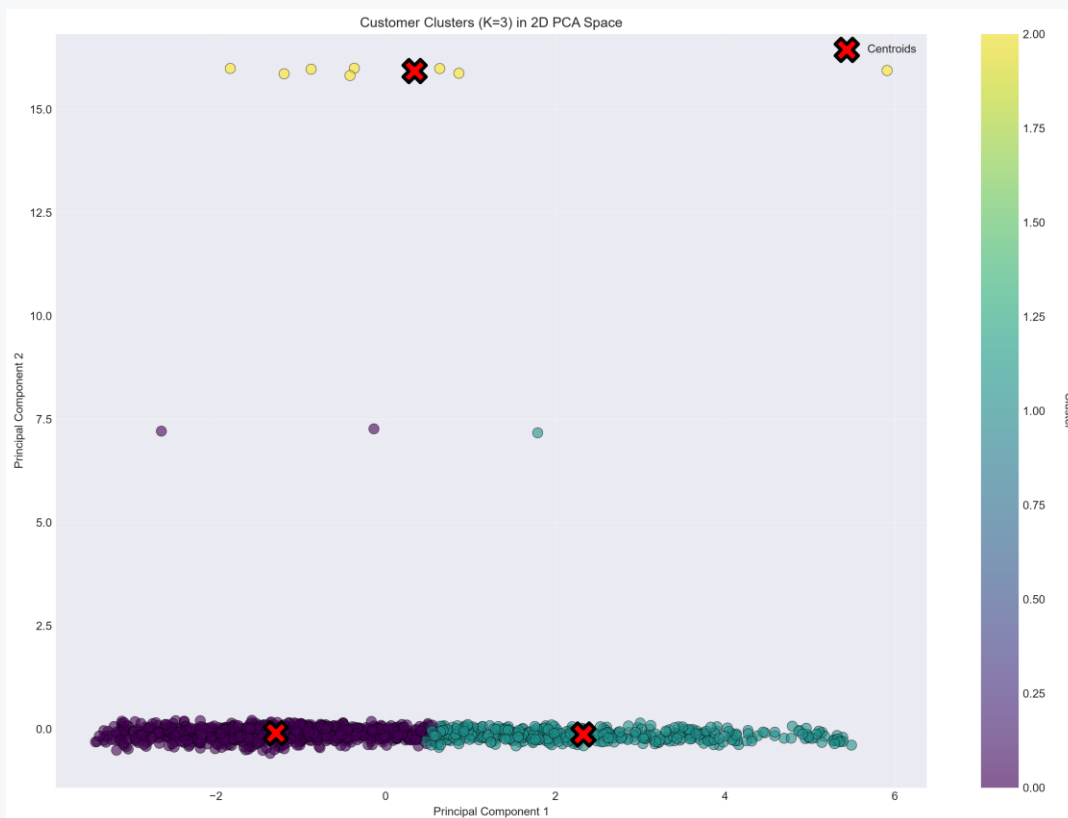


FIGURE 2. Customer clusters projected into 2D PCA space, with centroids marked.

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CLUSTER DIAGNOSTICS

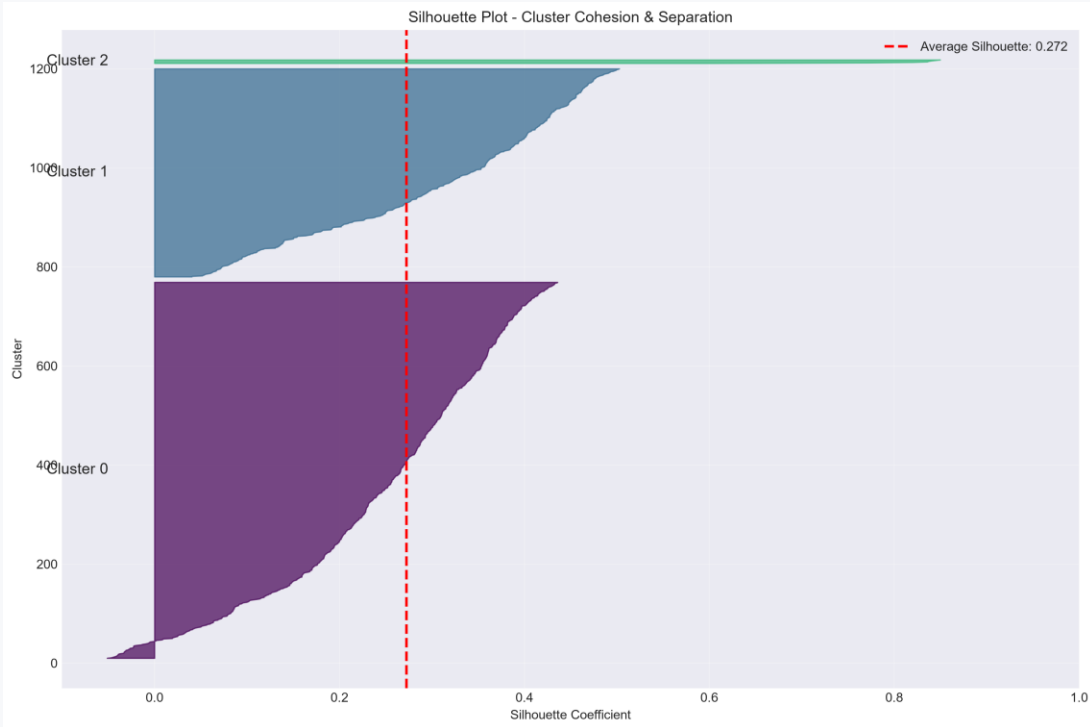


FIGURE 3. Silhouette plot showing cohesion and separation per cluster (average silhouette = 0.272).

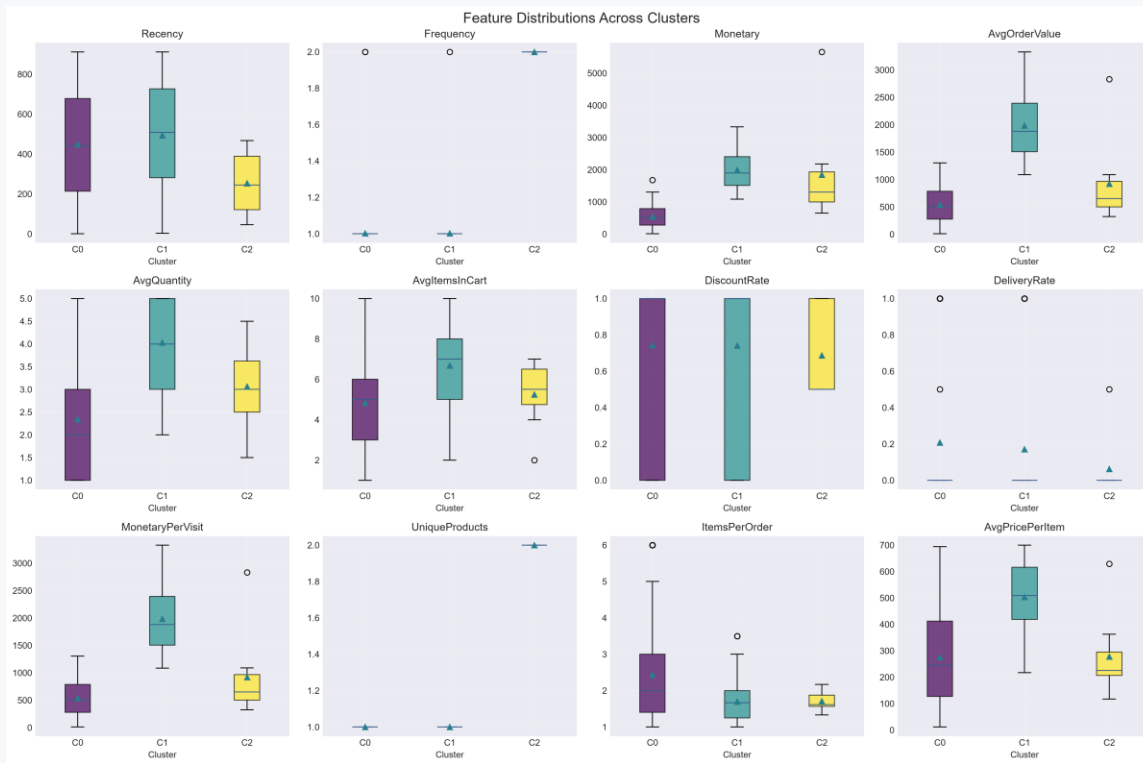


FIGURE 4. Distribution of each RFM and behavioral feature across the three clusters.

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CLUSTER PERSONAS

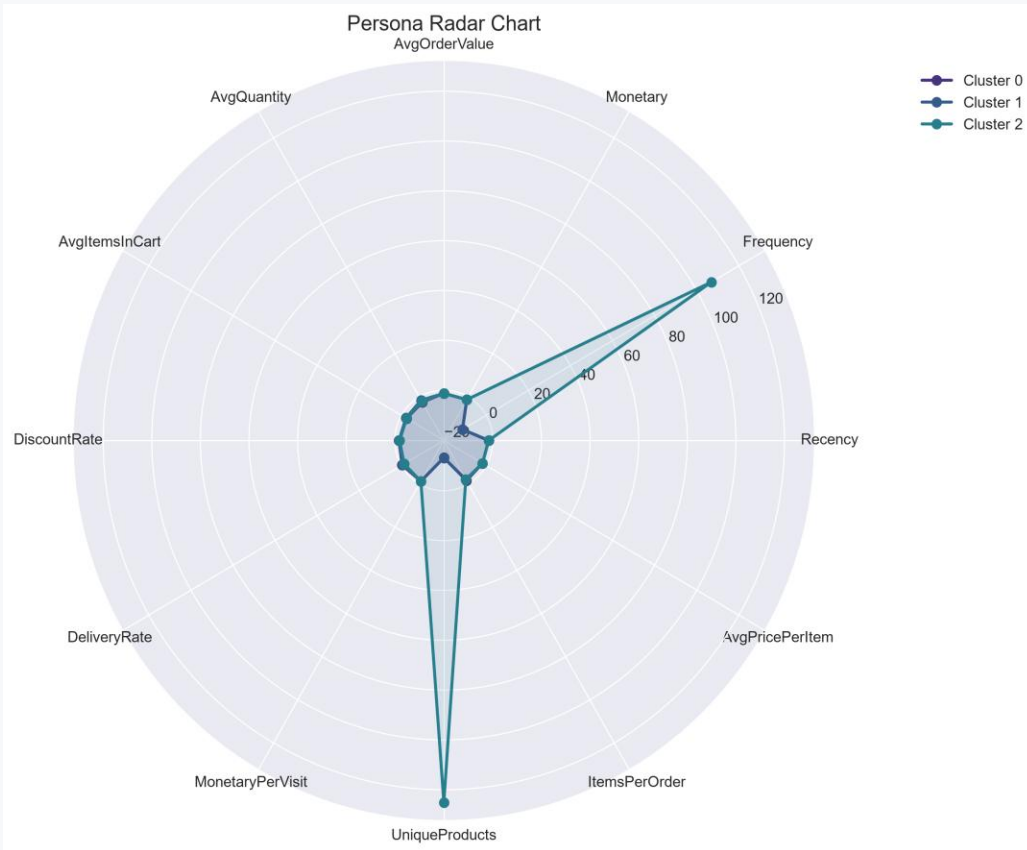


FIGURE 5. Persona radar chart comparing standardized feature profiles across clusters.

Customer Segment Summary

Cluster	Size	% of Total	Key Feature 1	Key Feature 2	Key Feature 3
Cluster 0	760	63.9%	Monetary: 545.8	MonetaryPerVisit: 548.5	AvgOrderValue: 548.5
Cluster 1	421	35.4%	MonetaryPerVisit: 1972.6	AvgOrderValue: 1972.6	Monetary: 1982.0
Cluster 2	8	0.7%	Frequency: 2.1	UniqueProducts: 1.9	Monetary: 1892.0

FIGURE 6. Customer segment summary table: size, share of base, and top distinguishing features.

Cluster Size, % of Base, and Defining Traits

Cluster 0 760 customers (63.9%). Lower Monetary (about 546) and AvgOrderValue. The largest, lower-value base segment.

Cluster	Size, % of Base, and Defining Traits
Cluster 1	421 customers (35.4%). High MonetaryPerVisit and AvgOrderValue (about 1,973). High-value repeat spenders.
Cluster 2	8 customers (0.7%). High Frequency (2.1) and UniqueProducts (1.9). A small, highly engaged niche group.

5 RECOMMENDED ACTIONS BY SEGMENT

- **High-value segment (Cluster 1)** — VIP loyalty program, early access, and personalized thank-you outreach.
- **Core, lower-value segment (Cluster 0)** — promotional testing and conversion-barrier analysis to migrate customers to higher-value tiers.
- **Small, highly engaged segment (Cluster 2)** — curated cross-sell recommendations and recognition given their multi-product, repeat engagement.

6 KEY TAKEAWAYS

- Three statistically distinct customer segments were identified from twelve RFM and behavioral features.
- PCA reduced dimensionality while retaining at least 95 percent of variance for stable clustering.
- An average silhouette score of 0.272 indicates moderate but usable cluster separation for business segmentation purposes.